Reading Financial Conditions in Units of Monetary Policy

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Abstract

How many points of monetary policy is a movement in financial conditions worth? The answer rests on two choices: how to weight asset prices, and the neutral level against which they are measured. I compare four calibrations of the weights on the 10-year yield, equity valuation and the dollar: market weights estimated from two measures of policy news in announcement windows, the spending weights of the FRB/US model, and the published Goldman Sachs weights. The futures-based market weights and the Goldman Sachs weights read a dollar appreciation relative to a fall in stock prices almost identically, while the two-year-yield market weights and FRB/US give the dollar more than twice the relative weight. Over 1996–2026 the four readings usually agree on the direction of a change in financial conditions and often disagree about its size: the highest and lowest readings differ by at least a quarter point of policy in about seven-tenths of months. Measuring yields and valuations relative to the neutral real rate adds about a point of policy to the 2008–09 tightening under every calibration, more than the spread across them.

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1 Introduction

How much should a rise in the dollar offset a rise in stock prices? And how tight is a given bond yield when the neutral interest rate falls? Every reading of financial conditions in points of monetary policy answers both questions, usually without saying so. In the year to January 1998 the dollar rose about 13 percent in real terms, equity valuations rose by about as much and the 10-year yield fell. Three of the calibrations compared in this paper read those changes as an easing of between 0.41 and 0.49 point. The fourth reads them as a tightening of 0.28 point. All four use the same three asset prices and the same policy unit. They differ in how much the dollar counts against the other two prices.

Policymakers read financial conditions in the units in which they set policy. In November 2023 the Chair noted that "Financial conditions have tightened significantly in recent months, driven by higher longer-term bond yields, among other factors" (Powell, 2023), and in March 2024 that "there are many different financial conditions indicators, and you can kind of, you know, see different answers to that question" (Powell, 2024). A reading in points of policy rests on two choices: the weights that combine asset prices, and the neutral level against which each asset price is measured. I hold everything else fixed. The asset prices are the 10-year Treasury yield, equity valuation and the dollar, with common transformations, and the unit is the usual response of the three prices to policy news that raises the two-year Treasury yield by one percentage point.

There are two ways to translate a movement in asset prices into that unit. One asks how large a monetary policy announcement would ordinarily produce the movement. The other asks how large a change in the policy rate would have a comparable effect on spending. I compare four calibrations of the two answers. Two are market-based: they weight asset prices by how they respond to monetary policy news in announcement windows, with the news measured either by a factor from interest-rate futures or by the change in the two-year Treasury yield. Two are spending-based: they weight asset prices by their effects on GDP, in the Federal Reserve's FRB/US model and in the published weights of the Goldman Sachs (GS) index. I call each a calibration, and the number it assigns to a movement a reading.

Figure 1 applies the four calibrations to two movements, a 5 percent fall in stock prices and a 4 percent appreciation of the dollar. The sizes are chosen so that the futures-based market calibration rates the two movements alike; the ratio of one in that calibration is set by the choice, not estimated. The published GS weights also rate them alike. The other two calibrations, one market-based and one spending-based, read the dollar appreciation as more than twice as tight as the fall in stock prices. Two indexes that both claim to measure financial conditions in units of policy can therefore differ by a factor of two on which of two ordinary movements is the larger tightening. And the split does not run between the market-based and the spending-based calibrations.

An asset earns weight in a market-based index by tracking policy news reliably. It earns weight in a spending-based index by moving spending. The calibrations agree exactly on a movement in the proportions that policy news produces and can differ only on the departure from those proportions (Section 2.3). Over 1996–2026 the four readings of annual changes usually agree on direction and

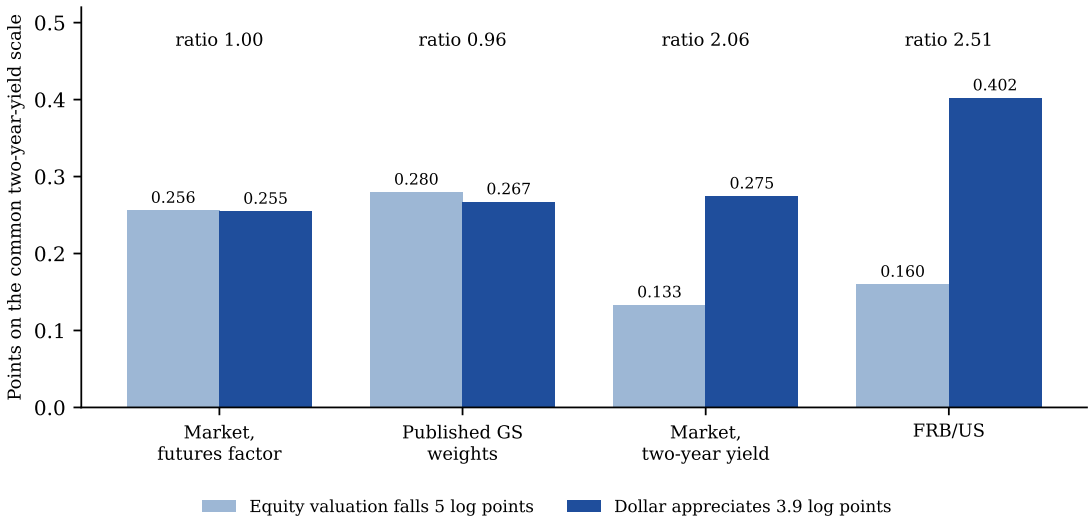


Figure 1: Two Tightenings Read by Four Calibrations

Notes: Readings of a 5 log point fall in equity valuation and a 3.9 log point appreciation of the real dollar, with the 10-year yield unchanged, under four calibrations on a common scale: each reads one on the usual response of the three asset prices to policy news that raises the two-year yield by one percentage point (Section 2.1). Market: weights from announcement-window responses, with policy news measured by the futures factor or by the change in the two-year yield. FRB/US: effects on GDP after four quarters of persistent shifts. Published GS weights: the U.S. weights on the 10-year yield, equities and the dollar reported by Caballero and Simsek (2026), applied to the same series, with the GS policy-rate and spread terms held fixed. The ratio above each pair is the dollar reading divided by the equity reading. The chosen movements give a ratio of about one under the futures-based calibration. Changing their relative sizes multiplies all four ratios by the same factor.

often disagree on size. In 2008–09, when equity valuations collapsed, the highest and lowest readings differed by 0.84 point of policy a year. Subsequent GDP growth offers little basis for choosing between the futures-based and FRB/US readings.

The neutral level matters as much. In 2008–09 the fall in the estimated r^*, the real interest rate consistent with output at potential, adds between 0.92 and 1.21 points to the tightening read by the four calibrations, more than the spread across them. Measuring yields and valuations relative to r^* removes one source of movement in neutral financial conditions. The resulting index tracks the gap to its own neutral level exactly, up to a constant, when the weighted neutral premiums are constant; this connects the paper to FCI*, the neutral level of financial conditions estimated by Caballero, Caravello and Simsek (2026).

1.1 Connection to the Literature

The weights and their meaning. The literature on monetary conditions indexes recognized that the weights on asset prices depend on the shock and on the dynamics, so that the policy meaning of a point is not invariant (Eika, Ericsson and Nymoen, 1996). Modern indexes differ mainly in the source of their weights (Hatzius et al., 2010): effects on output from a model (Dudley and Hatzius, 2000; Hatzius and Stehn, 2018; Ajello et al., 2023), principal components (Brave and Butters, 2011), or responses to policy news in event studies (Kuttner, 2001; Gürkaynak, Sack and Swanson, 2005; Bernanke and Kuttner, 2005; Nakamura and Steinsson, 2018; Bauer and Swanson, 2023*b*). I give that old point a precise form, a condition under which market-based and spending-based weights coincide, and measure how far four U.S. calibrations are from it. Whether announcement responses reflect policy or the central bank's information is debated (Nakamura and Steinsson, 2018); Bauer and Swanson (2023*a*) attribute most of the apparent information effect to the Federal Reserve's response to public news, and I use their orthogonalized surprise to check the market weights.

Comparisons of indexes. Comparisons of published indexes mix differences in weights with differences in assets, transformations and units. My design holds the assets, the transformations and the policy unit fixed across calibrations, so that a difference in readings is a difference in weights. The closest work translates indexes into policy units one at a time. Arnaut and Bauer (2024) estimate how an existing index responds to policy surprises, Sack and Swanson (2023) relate FCI-G, the Board's growth-impulse index, to policy-induced changes in the two-year yield, and Ajello et al. (2023) translate movements into a growth impulse with model multipliers. Bletzinger, Martorana and Mistak (2026) build a U.S. macro-finance index in units comparable to a policy rate and compare it with the GS index and FCI-G, finding divergences driven by equities and a sharp tightening in 2022–24.

Why the meaning of a point matters. Caballero, Caravello and Simsek (2024) study targeting an index of financial conditions in a closed economy, and Caballero and Simsek (2026) extend the analysis to an open economy in which domestic asset prices and the exchange rate both enter aggregate demand. That gives a reason to care which mix of asset prices delivers a given reading. My question comes one step earlier: when does an index built from policy news give those

markets the same relative importance as one built from spending effects? The same question arises for anyone who communicates about a desired path of financial conditions.

The neutral level. Estimates of r^* (Laubach and Williams, 2003; Holston, Laubach and Williams, 2017, 2023) are the standard benchmark for the stance of the policy rate, and indexes of financial conditions are usually read relative to their own history. Caballero, Caravello and Simsek (2026) define FCI*, the level of financial conditions consistent with output at potential, and estimate it for FCI-G in the manner of Laubach and Williams (2003). FCI* addresses the neutral level of the index as a whole and takes the weights of FCI-G as given. This paper takes r^* as given, measures each asset price relative to it, and shows how that relates to the gap to FCI*; the two choices are then measured on the same assets and the same scale.

2 What Does a Point of Financial Conditions Mean?

This section defines the two methods of translating a movement in asset prices into points of policy and shows when they give the same answer. The result is a simple identity that organizes the rest of the paper: two indexes can disagree only about the part of a movement that departs from the usual response to policy news.

2.1 Three Asset Prices, Two Methods, One Scale

I work with three asset prices throughout: the 10-year Treasury yield, equity valuation and the dollar. I use these three because every calibration in the comparison assigns them a weight and because their responses to monetary policy announcements can be measured in narrow windows. The 10-year yield is measured less expected inflation and less r^*, the real interest rate consistent with output at potential, as estimated by Laubach and Williams (2003). Equity valuation is the gap between the cyclically adjusted earnings yield and r^*, expressed in log points of the price-earnings ratio; a fall of one log point is a fall of about one percent in stock prices at given earnings, and I describe it that way below. Measuring yields and valuations relative to r^* keeps a decline in the neutral rate from registering as an easing (Section 3). The dollar is the log of the real broad dollar; it is a real exchange rate and is not measured relative to r^*. Appendix B gives the construction.

Two frequencies are involved, and the series differ between them. The responses to policy news are measured in narrow windows around announcements, where the S&P 500 and the DXY dollar index are available; the 10-year yield is the same series at both frequencies. The weights estimated from those responses are then applied, in the monthly history, to equity valuation and the real broad dollar, treating a one percent window return as one log point of valuation and the two dollar indexes as the same unit. All four calibrations are applied to the same monthly series. Differences among their readings therefore come from the weights and not from the data.

The first method starts from policy and asks what it does to markets. Around a Federal Open

Market Committee announcement, policy news of size s moves the three asset prices by

$$x = \psi s + u. \tag{1}$$

The vector ψ is the usual response of the three prices to one unit of policy news, the news that raises the two-year yield by one percentage point. The vector u collects everything else that moves the prices in the window, with covariance Σ. A market-based index reads a movement x as the amount of policy news that would ordinarily produce it.

The second method starts from markets and asks what they do to spending. Let g_h be the effects on GDP, h quarters ahead, of a unit movement in each asset price that follows a specified path:

$$\Delta y_h = g_h' x. \tag{2}$$

A spending-based index reads a movement x as the change in the policy rate that would have the same effect on GDP.

Neither reading is a claim about what caused the movement. A market-based reading of half a point says that the three prices moved as they would after policy news of that size; it does not say that policy news occurred. A spending-based reading of half a point says that the movement, if it lasts as assumed, will restrain GDP as much as a half-point rise in the policy rate. Both translate a financial movement into the units of policy, and each is useful for that reason: the first relates what markets did to what the central bank's own announcements usually do, and the second relates it to what the central bank is trying to achieve. An announcement can also reveal the central bank's view of the economy, so that the responses ψ mix policy and information (Nakamura and Steinsson, 2018); I measure the responses as markets deliver them and check the weights against a measure of policy news purged of public information (Section 4.1).

The methods have different units: points of the two-year yield for the first, points of the federal funds rate for the second, and index points for the published GS index. To compare weights I put every calibration on one scale. Each set of weights is divided by its own reading of the usual response ψ, so that every calibration reads one on it. I call this the common scale, and all readings below are on it. The two native units turn out to be nearly the same size: FRB/US reads the usual asset response to policy news that raises the two-year yield by one point as about one point on the funds rate (Section 4.2).

2.2 When Would the Two Methods Agree?

Take the case most favorable to agreement. Suppose that policy affects spending only through these three asset prices, at the horizon and for the paths that the spending calculation assumes. Would the assets that best reveal policy news then be the assets that matter most for spending?

The weights answer the question. The market weights are those of a regression that recovers the policy news from the three prices: an asset receives more weight the more it responds to policy news and the less it moves for other reasons. Formally they are generalized least squares weights.

On the common scale the two sets of weights are

$$w_M = \frac{\Sigma^{-1}\psi}{\psi'\Sigma^{-1}\psi}, \qquad w_{S,h} = \frac{g_h}{g_h'\psi}, \tag{3}$$

with M for market and S for spending. The two give the same reading for every possible movement in asset prices if and only if

$$g_h \propto \Sigma^{-1}\psi. \tag{4}$$

Condition (4) ties together two things that have no reason to be tied. The right-hand side is about signal quality: how strongly each asset responds to policy news, and how much it moves for other reasons. The left-hand side is about transmission: what a movement in each asset does to consumption, investment and trade. This is the precise form of the point that Eika, Ericsson and Nymoen (1996) made about monetary conditions indexes.

Two examples show how the two sides come apart. Suppose equities respond strongly to policy news but also move a great deal on earnings news that arrives in the same windows. Their market weight is then modest, even if a persistent fall in valuations has a large effect on consumption. Conversely, the dollar can respond to policy news so consistently that it is the best single indicator of the news, in an economy where trade is a small share of spending and the dollar matters little for GDP. A market-based index and a spending-based index can each be right about its own question and still disagree about a given movement. The same reasoning applies within each method: two market indexes that define policy news differently have different ψ and Σ, and two spending calibrations have different g_h. Whether the differences between methods exceed the differences within them is an empirical question, which Section 5 takes up.

2.3 Agreement in the History We Observe

Agreement for every possible movement is a demanding standard. What a policymaker needs to know is whether the readings agree for the movements that occur. Let x_t be the change in the three asset prices over some period, and $I_{M,t} = w_M' x_t$ its market reading. Split the movement into the part that policy news of that size would produce and a departure from it:

$$x_t = \psi I_{M,t} + v_t. \tag{5}$$

The departure v_t is observable given the weights and the usual response; computing it does not require knowing what moved markets. Every calibration reads the usual response as one point, so for any two calibrations j and k

$$I_{j,t} - I_{k,t} = (w_j - w_k)'v_t = \sum_a (w_{j,a} - w_{k,a})\, x_{a,t}, \tag{6}$$

where a runs over the 10-year yield, equities and the dollar. The first equality says that two calibrations can differ only on the departure from the usual response. The second is accounting:

each term is a difference in weights times the movement in one asset price, and the terms can reinforce or offset each other.

Three cases follow.

All calibrations agree when asset prices move in the usual proportions. Policy news that raises the two-year yield by one point raises the 10-year yield by about half a point, lowers stock prices by about 8 percent and raises the dollar by about 4 percent. A year with those movements is one point of tightening under every calibration, whether policy news, a change in risk appetite or developments abroad produced it.

The announcement of 28 January 2004, when the Committee dropped its statement that policy accommodation could be maintained for a "considerable period," comes close to that case. In the window the two-year yield rose 16.5 basis points, the 10-year yield 9.8 basis points and the dollar 1.0 percent, and stock prices fell 0.9 percent. Each weight converts one of these movements into points of policy, and the reading is the sum. Under the futures-based weights the rise in the 10-year yield contributes 0.06 point, the fall in stock prices 0.05 and the rise in the dollar 0.07, a tightening of 0.17 point. The four calibrations read the announcement as a tightening of between 0.17 and 0.19 point, because the three prices moved roughly as policy news moves them.

Actual announcements come close to this benchmark without reaching it. Estimating the weights with each announcement held out and reading that announcement's window movement, the mean absolute difference between the futures-based and FRB/US readings is 0.19 of the mean absolute market reading. For 12-month changes in the monthly history it is 0.33. The 90 percent intervals run from 0.09 to 0.37 and from 0.17 to 0.63. Asset prices depart from the usual proportions even on announcement days, and further over a year.

Calibrations disagree when one asset price moves far and they weight it differently. A year in which stock prices fall 16 percent and nothing else moves is not a multiple of the usual pattern. Its departure is large, and calibrations that weight equities differently read it differently. The intermeeting cut of 3 January 2001 is an example within a single window. Stock prices rose 4.4 percent, about four times the usual response to news of that size, and the dollar rose 0.3 percent. The readings run from an easing of 0.08 point under the two-year-yield weights, which give equities the least weight, to 0.22 point under the published GS weights.

Calibrations with different weights can still agree, because the terms of equation (6) can offset. If stock prices fall and the dollar rises, a calibration that gives equities less weight and the dollar more can arrive at the same reading as one that does the reverse. As a result, two indexes can track each other for years without embodying the same economics, and agreement in one period says little about agreement in the next. What matters is which asset prices move, and in which combinations.

3 Measured Relative to What? The Neutral Level

A reading in points of policy says how far financial conditions have moved, or where they stand, relative to something. For changes over a year the reference matters little, because the neutral

level rarely moves much within a year. For levels, and in the years when the neutral level moves quickly, it matters as much as the weights, and in 2008–09 more. This section sets out the three steps from nominal measures to measures relative to r^*, relates the last step to FCI*, and compares the readings with the published estimate of FCI*.

3.1 From Nominal Yields to Yields Relative to r^*

The first step removes expected inflation from the 10-year yield. The second subtracts r^* from the real yield and from the cyclically adjusted earnings yield, which is the real return that equity valuations imply. The dollar is a real exchange rate throughout and is not adjusted for r^*. Appendix B gives the series.

The 10-year yield shows what each step does. It averaged 11.5 percent in 1980 and 4.0 percent in 2023. Less expected inflation over the next ten years, the two yields were nearly equal, at 1.7 and 1.8 percent. Less r^*, the 1980 yield was 1.5 points below neutral and the 2023 yield 0.2 point above it. A 10-year yield nearly three times as high was the easier of the two.

The index as a whole adds equity valuations and the dollar, and they change the comparison. Under the futures-based weights, the index reads 1980 as 9.4 points tighter than 2023 in nominal terms, 4.7 points tighter in real terms and 3.3 points tighter relative to r^*. What remains is equity valuation: stock prices were low relative to earnings in 1980, with a cyclically adjusted price-earnings ratio of about 8.8 against 29.5 in 2023, while relative to r^* the 10-year yield and the dollar both read 2023 as the tighter year.

3.2 Relative to r^* and Relative to FCI*

Caballero, Caravello and Simsek (2026) define FCI* as the level of a financial conditions index consistent with output at potential. They work with FCI-G, which measures the effect of asset prices and interest rates on expected output growth, and they infer FCI* from the relation between the gap of FCI-G to FCI* and the output gap, with a Kalman filter in the manner of Laubach and Williams (2003). Because the index already carries the demand effects of asset prices, FCI* reflects what the economy requires rather than movements in asset prices themselves. In their estimates the r^* of Holston, Laubach and Williams (2023) moves with the equity risk premium and FCI* does not.

An estimate of FCI* requires a model of the output gap and of how financial conditions affect it. The readings in this paper are built from asset prices and fixed weights alone, so I do not estimate a neutral level of the index. Measuring each asset price relative to r^* removes one source of movement in the neutral level. To see the connection, write the neutral level of each asset price as r^* plus a neutral premium:

$$x_t^* = \iota\, r_t^* + c_t, \tag{7}$$

where ι gives the loading of each asset on r^* in the units of the index (one for the 10-year yield, a conversion factor for equity valuation and zero for the dollar) and c_t collects the neutral term premium, the neutral equity premium and the neutral level of the dollar. The gap of a reading to

Table 1: Readings and the Gap to FCI*

	Level, r^* fixed	Level, r^*	4-quarter change, r^* fixed	4-quarter change, r^*
Market, futures factor	0.48	0.69	0.82	0.81
Market, two-year yield	0.22	0.64	0.64	0.75
FRB/US	0.32	0.69	0.78	0.83
Published GS weights	0.54	0.69	0.84	0.82

Notes: Correlation of each calibration's reading (quarterly mean; common scale) with the gap between FCI-G and FCI* as published by Caballero, Caravello and Simsek (2026), 1990Q2 to 2024Q4. r^*: yields and valuations relative to the Laubach and Williams (2003) estimate, as in the rest of the paper; r^* fixed: the same series with r^* held at its 1973–2026 mean. FCI-G is the 3-year-lookback version and FCI* the one-sided estimate.

its neutral level is

$$w'(x_t - x_t^*) = w'(x_t - \iota\, r_t^*) - w'c_t. \tag{8}$$

The first term on the right is the reading relative to r^* used throughout this paper. Subtracting r^* removes the part of the neutral level that moves with the neutral real rate. When the neutral premiums are constant, $w'c_t$ is a constant, and the demeaned reading relative to r^* is the gap to neutral. The two differ by the weighted movement in neutral premiums. Holding r^* at its sample mean with everything else unchanged shows how much the first term owes to movements in r^*: in 2008–09 those movements add between 0.92 and 1.21 points to the tightening read by the four calibrations (Section 5.1).

3.3 The Readings and the Published FCI*

Caballero, Caravello and Simsek (2026) publish their one-sided FCI* with FCI-G for 1990Q2 to 2024Q4. Table 1 correlates each calibration's quarterly reading with the gap between FCI-G and FCI*. The two indexes differ in assets, transformations and weights, so the correlation measures common movement, not identity.

In levels, measuring relative to r^* brings every calibration closer to the FCI* gap. With r^* held at its sample mean, the correlations run from 0.22 to 0.54; relative to r^* they run from 0.64 to 0.69. The correlations, which differ widely across calibrations with r^* held fixed, nearly coincide relative to r^*. A higher correlation does not show that either series measures the true stance; it shows that the two neutral benchmarks move together. In 4-quarter changes the correlations are between 0.64 and 0.84 with or without r^*, because the neutral rate moves little within a year. The estimated r^* and FCI* move together, with a correlation of 0.37 in levels, and both register the 2008–09 crisis as a period in which a given set of asset prices was tighter relative to neutral.

4 Why Do the Four Calibrations Weight the Assets Differently?

The four calibrations differ mainly on one question: how much a movement in the dollar counts relative to a movement in stock prices. Table 2 gives the weights on the common scale. The

Table 2: Four Sets of Weights on One Scale

	Market, futures factor	Published GS weights	Market, two-year yield	FRB/US
10-year yield (per percentage point)	0.609 [0.147, 1.114]	0.515	0.926 [0.476, 1.328]	0.590
Equity valuation (per log point)	−0.051 [−0.068, −0.030]	−0.056	−0.027 [−0.046, −0.011]	−0.032
Dollar (per log point)	0.065 [0.021, 0.112]	0.069	0.070 [0.033, 0.110]	0.103

Notes: Weights on the three assets relative to r^*, each set normalized to read one on the response of the three assets to one unit of futures-factor policy news (the common two-year-yield scale). Market weights: generalized least squares on 253 announcement windows, with policy news measured by the futures factor or by the change in the two-year yield; brackets are 90 percent intervals over 5,000 bootstrap draws of the announcements (for the two-year-yield weights, of the weights on their own signal, which differ from the entries by a factor of 0.999). FRB/US: effects on GDP after four quarters of persistent shifts. Published GS weights: the U.S. weights on the 10-year yield, equities and the dollar, with the GS policy-rate and spread terms held fixed. The FRB/US and GS weights are not estimated here.

futures-based market weights and the published GS weights are close on all three assets. The market weights based on the two-year yield put more weight on the 10-year yield and about half as much on equities. FRB/US puts the most weight on the dollar. This section traces those differences to two sources: the definition of policy news on the market side, and the channels of spending on the spending side.

4.1 The Market Side: What Counts as Policy News

The market weights come from the 253 announcements from January 1996 to September 2026 in the San Francisco Fed's U.S. Monetary Policy Event-Study Database (Acosta et al., 2025). In a narrow window around each announcement I measure the changes in the 10-year yield, the S&P 500 and the dollar, and a measure of the policy news. The weights recover the news from the three changes, so they depend on what the researcher counts as news. The choice looks like a technical detail. It turns out to change the weight on equities by half.

Two measures are natural. The first is the factor of Nakamura and Steinsson (2018), the common component of the changes in five interest-rate futures that span the next year or so of expected policy. The second is the change in the two-year Treasury yield in the same windows, which is the unit of the index. The two share most of their variation; their correlation is 0.90. But they are not the same. Write the two-year change as the futures factor plus a residual. The residual accounts for about a fifth of the variance of the two-year change.

Figure 2 shows that the two parts move different assets. The futures factor moves all three: yields up, stock prices down, the dollar up. The residual moves the 10-year yield and the dollar at least as much as the factor does, and it does not move stock prices. That announcements carry more than one dimension of news is well established: Gürkaynak, Sack and Swanson (2005) separate news about the current policy rate from news about its path, and Boehm and Kroner (2025) find a dimension that moves equities and exchange rates but not yields. In these data, the component of the two-year change left unexplained by futures is associated with movements in long yields and the dollar, with little estimated equity response.

The same movement in yields and the dollar is noise under one definition of policy news and

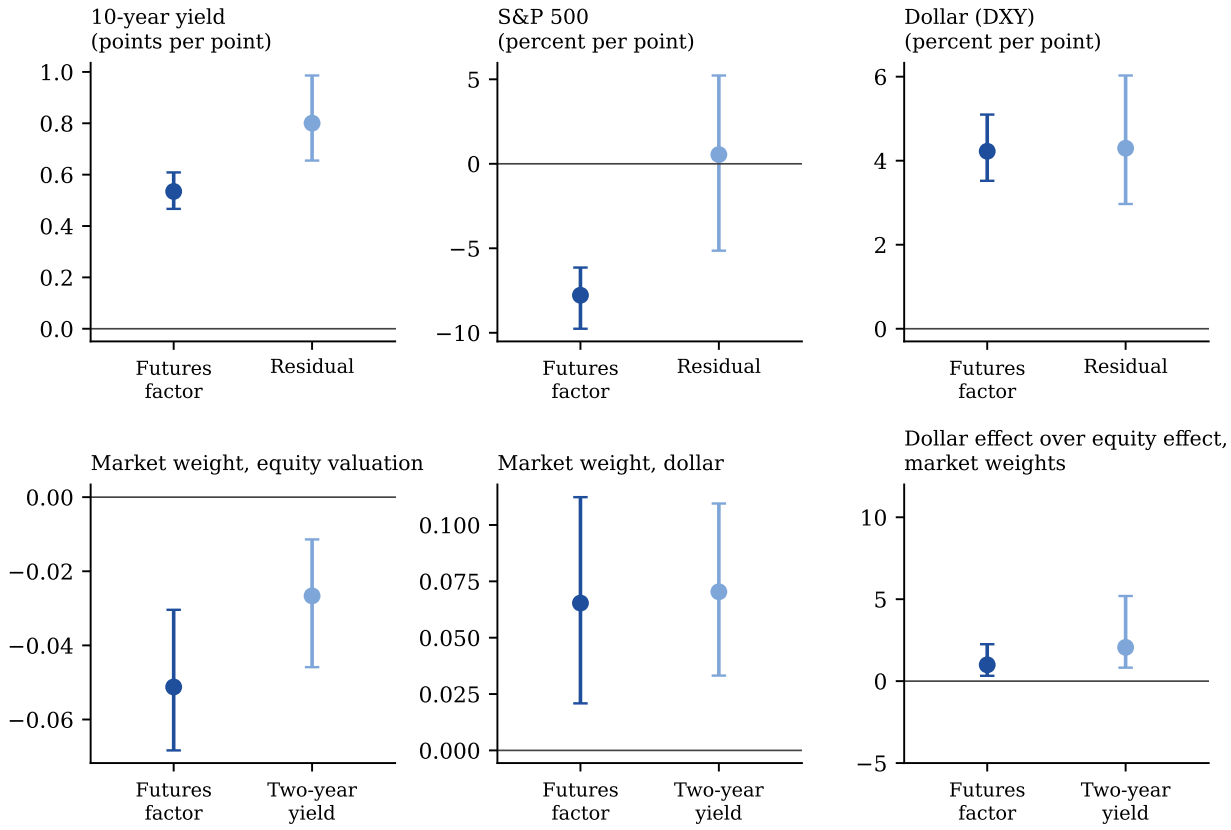


Figure 2: What the Two Measures of Policy News Capture

Notes: Top: responses of the 10-year yield (points), the S&P 500 and the dollar (DXY) (percent) in the announcement windows to one point of the futures factor and to one point of the residual, the part of the two-year yield change that is orthogonal to the factor; the factor is scaled to the two-year change. Bottom: the market weight on equity valuation and on the dollar, and the market's dollar-over-equity ratio, when policy news is measured by the futures factor and by the two-year yield. Bars are 90 percent intervals from 5,000 bootstrap draws of the announcements, drawn in common for the two measures.

signal under the other, and equation (3) turns that into different weights. When policy news is the futures factor, the residual is part of u: it moves the 10-year yield and the dollar for reasons unrelated to the news and makes them less reliable indicators of it. Equities, which do not respond to the residual, are the cleaner signal and receive a large weight. When policy news is the two-year yield, the residual is part of the news. The 10-year yield and the dollar now track the news closely, and equities, which respond to only part of it, are the less informative asset. The weight on equity valuation falls from -0.051 to -0.027, and the dollar's weight relative to equities roughly doubles. Comparing the two definitions on the same bootstrap draws of the announcements, the 90 percent interval of that multiple runs from 0.98 to 5.47.

Which measure is right depends on what the residual is. If it is policy news about horizons beyond those the futures cover, the two-year yield is the better measure of the news a market-based index should recover. If it is a movement in term premiums around announcements, or measurement error in a single yield, the futures factor is. The announcement responses cannot tell these apart, and the choice moves the answer to the question of Figure 1 from one to two.

A third measure addresses the information in announcements. An announcement can reveal the Federal Reserve's view of the economy as well as its policy (Nakamura and Steinsson, 2018). Bauer and Swanson (2023*a*) show that much of what looks like such information is the Federal Reserve's response to economic news that was public before the announcement, and Bauer and Swanson (2023*b*) construct a surprise orthogonalized to that news. Measuring policy news with their surprise, on the 200 announcements through 2019 that it covers, gives weights of 0.542 on the 10-year yield, -0.070 on equity valuation and 0.040 on the dollar (Table C.2). Equities gain weight relative to the dollar: the dollar-over-equity ratio of Figure 1 is 0.44, against 0.68 for the futures factor on the same announcements. The readings of history move by less than the distance between the futures-based and FRB/US calibrations, and FRB/US remains the calibration that gives the dollar the most weight.

4.2 The Spending Side: What Happens After Asset Prices Move

The spending weights answer a different question, and FRB/US lets a reader see the answer take shape (Brayton, Laubach and Reifschneider, 2014). Figure 3 traces the two movements of Figure 1 through the model.[1]

The fall in stock prices works through domestic demand. Lower equity wealth reduces consumption, which supplies 92 percent of the fall in GDP after four quarters, and a higher cost of capital reduces business investment. Weaker demand also reduces imports, so net exports offset part of the decline. The appreciation of the dollar works through trade. Net exports account for 94 percent of the fall in GDP, mostly through lower exports, and domestic demand hardly moves. The two channels work at different speeds. An appreciation changes the relative price of U.S. and

[1] Each asset price shifts persistently while the policy rate, the yield curve, credit spreads and the other asset price follow their baseline paths. The equity experiment raises the required return on equity, so that household stock wealth falls with valuation.

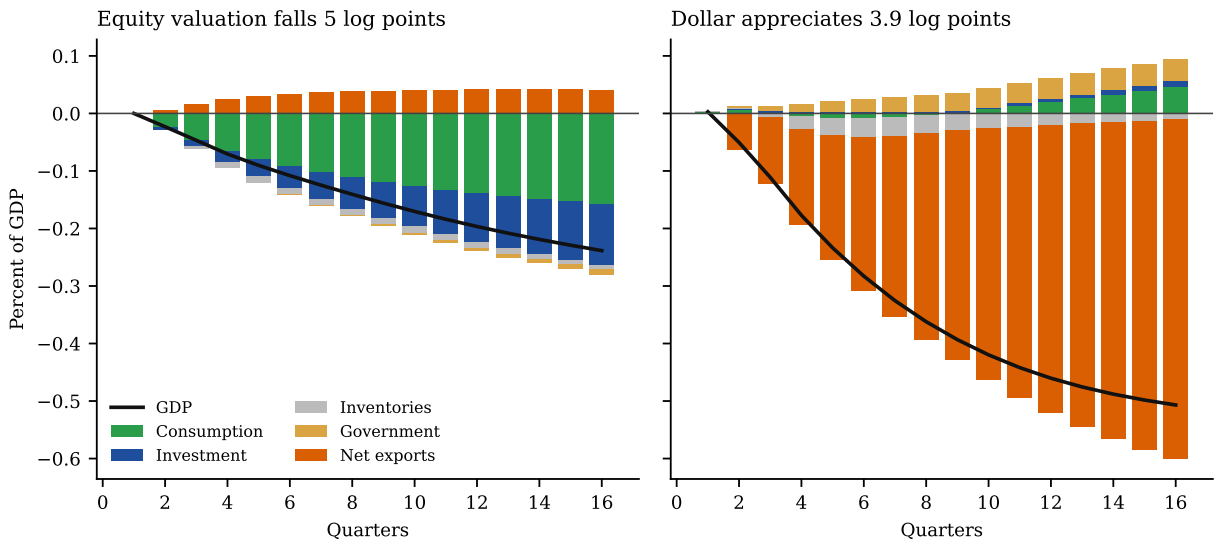


Figure 3: How the Two Movements Reach GDP in FRB/US

Notes: Percent deviation of real GDP from baseline and contributions by expenditure component, persistent shift in the variable, other financial variables held at baseline, policy rule inactive. Investment is residential and business fixed investment; net exports are exports less imports. The figure uses the Board's FRB/US package with the data vintage of October 2026.

foreign goods at once, and exports and imports respond within a few quarters: after four quarters the dollar has delivered 30 percent of its effect on GDP over eight years, and after two years 62 percent. A loss of wealth reaches consumption more slowly, as households adjust spending plans that look many years ahead: after four quarters the fall in stock prices has delivered 21 percent of its eight-year effect, and after two years 41 percent. A spending weight measured at four quarters therefore catches more of the trade channel than of the wealth channel. That is why FRB/US counts the dollar for more. The multipliers of FCI-G, which descend from FRB/US, point the same way: per percent, its dollar multiplier is 2.25 times its equity multiplier, against 3.22 in FRB/US (Ajello et al., 2023).

How long a movement lasts matters for a spending calculation in a way that it does not for a market calculation, and the weights in Table 2 assume that movements persist. Under that assumption FRB/US reads the usual response to policy news as 0.954 points on the funds rate; if the movements instead decay with a half-life of four quarters, it reads 0.82. Persistence therefore matters for how much tightening a movement represents and little for which asset counts more: with a common half-life of four quarters, the model's dollar effect is 2.55 times its equity effect for the movements of Figure 1, against 1.00 for the market weights. The conversion says how FRB/US values the mix of asset movements observed around announcements. A shock to the model's own policy rule produces a different mix, with more movement in equities and less in the dollar.

4.3 A Second Spending Calibration

If FRB/US were the only spending calibration, a reader might conclude that spending-based indexes weight the dollar more heavily than market-based ones. The published GS weights show that this does not follow. Like FRB/US, they weight each variable by its estimated effect on GDP growth over the following year (Dudley and Hatzius, 2000; Hatzius and Stehn, 2018). Caballero and Simsek (2026) report the current U.S. weights and illustrate them with an example close to Figure 1: a 20 percent fall in equity prices and a 17 percent appreciation of the currency each tighten the index by about one point.[2] The published weights apply to variables in different units; Table 2 converts them to the units used here and to the common scale. On that scale they are close to the futures-based market weights on all three assets.

A spending calibration can therefore resemble a market calibration closely, and the two spending calibrations differ from each other by more. Relative to equities, FRB/US gives the dollar about two and a half times the weight that the published GS weights do. The two calibrations disagree about how much a movement in the dollar does to GDP within a year, relative to a movement in stock prices.

The four calibrations thus differ for three reasons. The market weights depend on what counts as policy news. The FRB/US weights reflect a trade channel that acts faster than the wealth channel. The GS weights reflect a different balance between the two.

5 When Does the Choice Change the Reading of History?

Differences in weights matter only if asset prices move in ways that expose them. This section asks how often they have, and whether the differences affect the direction of the reading or only its size. I apply the four calibrations to the change over the preceding 12 months in the three asset prices, in each of the 369 months from January 1996 to September 2026. Corporate spreads, which the GS index and the full FRB/US calculation include, are held fixed.

History answers three questions that the identity of Section 2.3 raises. How far apart can the readings be when the direction is clear? When can the sign of the reading reverse? And how can different weights produce nearly identical readings? Figure 4 shows the four readings in the three episodes that answer them: 2008–09, early 1998 and 2022–24.

5.1 2008–09: Agreement on Direction, Disagreement on Size

Equity valuations roughly halved between 2007 and March 2009, while the dollar changed little on average. One asset dominated the movement, so one difference in weights dominates equation (6). The futures-based market weights and the GS weights on equities are between one and a half and

[2]The U.S. weights are 4.4 on the policy rate, 45.1 on the 10-year yield, 39.6 on a corporate spread, 4.9 on an equity variable (prices scaled by a 10-year average of earnings) and 6.0 on a trade-weighted dollar (Caballero and Simsek, 2026). I apply the weights on the 10-year yield, equities and the dollar to the series used for the other calibrations and hold the policy-rate and spread terms fixed. The result is the published GS weights on the common assets, not the GS index.

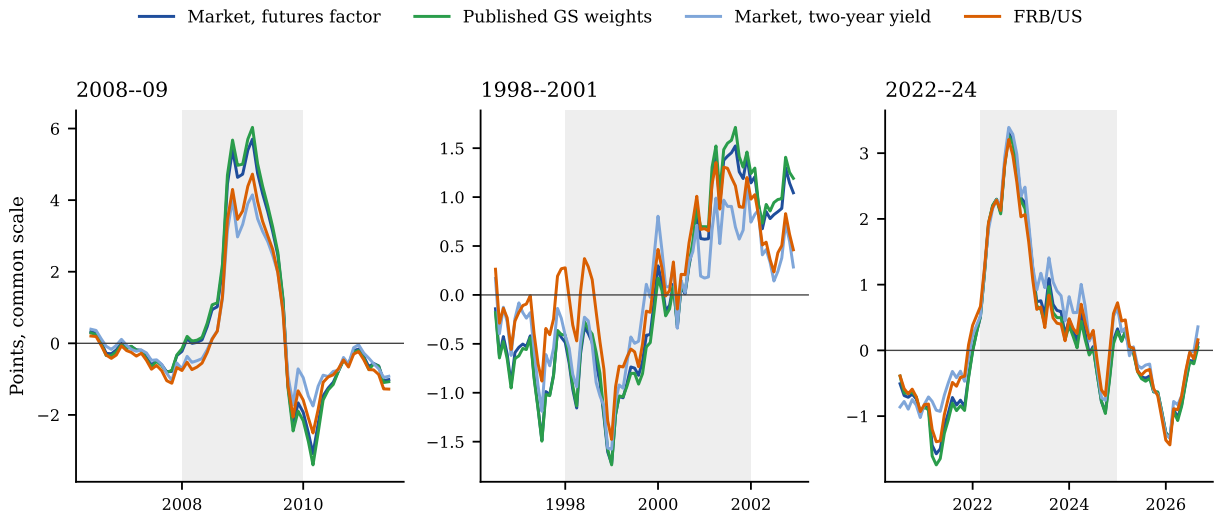


Figure 4: Four Readings, Month by Month

Notes: Twelve-month change in financial conditions under each calibration, in points on the common scale. The shaded months are January 2008 to December 2009, January 1998 to December 2001 and March 2022 to December 2024. Corporate spreads are held fixed.

two times the other two (Table 2). As a result, the futures-based and GS calibrations read the two years as about 2.05 points of tightening a year, and the two-year-yield and FRB/US calibrations as about 1.40 (Table 4). At the widest the highest and lowest readings were 2.00 points apart.

Every calibration says that financial conditions tightened sharply. They differ by more than half a point of policy on how sharply. A policymaker describing the tightening in these three markets would have quoted a figure more than half a point higher using the futures-based or GS weights than using the two-year-yield or FRB/US weights, and the direction of the movement did nothing to reveal which. The four lines in Figure 4 nevertheless rise and fall together; over the full sample the readings are correlated between 0.93 and 1.00. High correlation is a statement about direction and timing, and it is compatible with large differences in size. This is consistent with Bletzinger, Martorana and Mistak (2026), who find that the divergences among published U.S. indexes are driven by equities.

How conditions are measured relative to neutral matters as much for this episode as the choice of weights. The estimated r^* fell during the crisis, and measuring yields and valuations relative to it counts that fall as a tightening. Allowing r^* to vary adds 0.92 to 1.21 points to the average 2008–09 reading across the four calibrations, more than the 0.84-point spread between the highest and lowest readings. The disagreement over weights survives that large change in the estimated severity of the episode. With r^* held at its average, the futures-based and GS readings remain the two highest, and the four readings still span 0.75 point (Section 5.4 describes the calculation).

Table 3: January 1998: One Movement, Opposite Readings

	Change over the year	Market, futures factor	Published GS weights	Market, two-year yield	FRB/US
10-year yield	−1.0	−0.63	−0.53	−0.96	−0.61
Equity valuation	13.5	−0.69	−0.76	−0.36	−0.43
Dollar	12.8	0.84	0.88	0.90	1.32
Reading		−0.49	−0.41	−0.42	0.28

Notes: Change in each asset price over the 12 months to January 1998 (10-year yield relative to r^* in percentage points; equity valuation and the real dollar in log points) and its contribution to each reading (weight times change, points on the common scale). A positive contribution is a tightening.

5.2 Early 1998: Offsetting Movements and Opposite Signs

The second question is when the sign can reverse. In the year to January 1998, during the Asian financial crisis, the dollar appreciated by about 13 percent in real terms, equity valuations rose by about 13 percent and the 10-year yield fell by a point relative to r^*. The appreciation tightened conditions; the other two movements eased them.

The sign of the reading then depends on the weights, as Table 3 shows. Under the futures-based weights, the appreciation adds 0.84 point of tightening and the other two movements subtract more, for a net easing of half a point. Under the FRB/US weights the same appreciation adds 1.32 points, because FRB/US gives the dollar the most weight, and the reading is a tightening of a quarter point. The other two calibrations read an easing. An analyst watching household wealth and borrowing costs and an analyst watching the trade balance could have looked at the same year and given opposite accounts of it, and the four calibrations encode those two views in different proportions. The same configuration returned briefly in early 2015, when the dollar and stock prices again rose together, with the same split.

Opposite signs arise when asset movements push financial conditions in opposite directions and the calibrations differ on the asset that moved most. Opposing movements of that size are unusual, which is why such months are rare.

5.3 2022–24: Different Weights, the Same Reading

The third case is the one that equation (6) predicts and intuition resists. From March 2022 to December 2024 all three asset prices tightened conditions. The 10-year yield rose, equity valuations fell by about 5 percent a year on average and the dollar appreciated by about 3 percent a year. The four readings average about one point of tightening, and the futures-based and FRB/US readings are nearly identical.

They agree for a reason that has nothing to do with similar weights. Relative to the futures-based reading, FRB/US assigns 0.10 point less tightening to the fall in stock prices, because it gives equities less weight, and 0.11 point more to the appreciation, because it gives the dollar more. The two terms of equation (6) cancel. The same two differences in weights that separated the readings by more than half a point in 2008–09 and reversed their signs in early 1998 left them nearly equal

Table 4: Four Readings of History

	Relative to r^*	90% interval	Real, no r^* adjustment	Nominal
Mean absolute distance between two readings (points)				
Market, futures factor vs Published GS weights	0.07	[0.06, 0.44]	0.08	0.10
Market, futures factor vs FRB/US	0.26	[0.12, 0.50]	0.25	0.26
Market, two-year yield vs FRB/US	0.23	[0.08, 0.51]	0.25	0.32
Market, futures factor vs Market, two-year yield	0.28	[0.13, 0.42]	0.35	0.39
FRB/US vs Published GS weights	0.28	[0.24, 0.32]	0.31	0.32
Market, two-year yield vs Published GS weights	0.34	[0.16, 0.60]	0.43	0.49
Judgments across the four readings, percent of 369 months				
All four the same sign	81.6	[70.2, 82.9]	72.6	73.4
Materially opposing readings, threshold 0.25	2.2	[1.6, 8.1]	4.3	7.0
Materially opposing readings, threshold 0.10	10.3	[8.4, 19.8]	17.1	16.8
Largest less smallest reading at least 0.25	70.5	[58.5, 88.9]	75.1	78.3
Average reading, January 2008 to December 2009 (points)				
Market, futures factor	2.05		1.00	0.88
Published GS weights	2.17		1.15	1.06
Market, two-year yield	1.33		0.31	0.10
FRB/US	1.40		0.58	0.50

Notes: Twelve-month changes in the 10-year yield, equity valuation and the dollar, monthly, January 1996 to September 2026 (369 overlapping months), read with the four sets of weights of Table 2, in points on the common scale. Each column applies all four sets, unchanged, to the same series. Relative to r^*: the series of the paper. Real, no r^* adjustment: the real 10-year yield, 100 times the log of the cyclically adjusted price-earnings ratio (the documented Goldman Sachs equity concept) and the real dollar. Nominal: the nominal 10-year yield, the same equity measure and the nominal broad dollar, the nearest publicly reproducible Goldman Sachs inputs. Materially opposing: at least one reading at least the threshold above zero and one at least the threshold below. Intervals: 5th and 95th percentiles over 5,000 bootstrap draws of the announcements, for the series of the paper; they reflect the estimation of the market weights, conditional on the observed history and on the FRB/US and Goldman Sachs weights. Corporate spreads are held fixed.

here, because stock prices and the dollar both moved in the tightening direction by amounts that happened to balance. Month by month the cancellation was rougher: the FRB/US reading less the futures-based reading ranged from −0.32 to 0.40 point within the episode. Near-equality of two readings is therefore not evidence that two indexes embody the same economics.

5.4 How Representative Are the Episodes?

The three episodes were chosen to illustrate. Table 4 asks how often each case occurs across all 369 months.

Distances. The closest pair of point estimates crosses the two methods: the futures-based market reading and the published GS weights differ by 0.07 point on average, against 0.26 to 0.28 for the futures-based reading against FRB/US and against the other market calibration. The widest pairs include both pairs within a method. The same pair is closest in every set of series in Table 4. Its lead over the next-closest pair is not statistically distinguishable from zero: on common bootstrap draws the 90 percent interval runs from −0.16 to 0.28 point for the series of the paper. Switching between the two market calibrations, or between the two spending calibrations, changes the reading

by about as much on average as switching from the futures-based calibration to FRB/US.

Direction. In the 369 overlapping annual changes, the four readings share a sign about four-fifths of the time (81.6 percent). In 8 months they lie at least a quarter point on opposite sides of zero. Six of them, in late 1997 and 1998 and in early 2015, are months in which FRB/US alone read a tightening; the other two, in 2003 and 2008, came after falls of more than 20 percent in equity valuations. At a threshold of a tenth of a point the share is 10.3 percent.

Size. The highest and lowest of the four readings differ by at least a quarter point about seven-tenths of the time (70.5 percent). Disagreement about the size of a movement is the normal state of affairs; disagreement about its direction is the exception.

What the input series change. Published indexes use nominal or real series and do not adjust for the neutral rate. The last two columns of Table 4 ask whether the conclusions depend on that choice. They apply the same four sets of weights to different series. The paper's equity variable starts from the gap between the cyclically adjusted earnings yield and r^* and converts it to valuation units at a reference earnings yield; the second column instead uses the log of the cyclically adjusted price-earnings ratio directly, together with the real 10-year yield without the subtraction of r^*. The third column also replaces the real yield and the real dollar with their nominal counterparts, the nearest public versions of the GS inputs.

The ranking of the calibrations survives: the futures-based and GS readings remain the closest pair, and the distances within a method remain at least as large as the distance between methods. The readings themselves change. Measuring yields and valuations relative to r^* counts a fall in the neutral rate as a tightening, because a given yield or earnings yield is then higher relative to neutral. The neutral rate fell during the crisis, so the paper's series register more tightening in 2008–09 than series without the adjustment, and under the alternative series the crisis reading is roughly half as large or less under every calibration. The columns change more than r^*, so this comparison is the combined effect of the inputs. Holding r^* at its 1973–2026 average while keeping every other feature of the paper's series isolates its role. The 2008–09 readings then come out close to the second column, so the movement in r^* accounts for most of the difference (Section 5.1). The readings also split more often without the adjustment: all four share a sign in 73.4 percent of months. How a movement is measured relative to neutral matters for the level of every reading and for how often the readings split, and much less for which calibrations resemble each other. That echoes Kiley (2020), where the transformations of financial variables drive which variables an index selects, and Aramonte, Rosen and Schindler (2013), where comparisons among indexes depend on the crisis in the sample.

6 Can Subsequent Activity Distinguish the Readings?

6.1 The Question

If the readings differ, a natural response is to ask the data which is right. Consider two years in which the futures-based market reading tightened by the same amount. In one, the dollar accounted

for more of the tightening and stock prices for less, so that the FRB/US reading was higher. Did activity subsequently weaken more in that year? If the market reading is a sufficient summary of the three asset prices for predicting activity, the answer is no. If the FRB/US reading is sufficient, the answer is yes, and by a known amount.

The regression is

$$y_{t+h} - y_t = \sum_\ell \rho_\ell y_{t-\ell} + \beta I_{M,t} + \gamma D_t + \varepsilon_{t+h}, \tag{9}$$

where $I_{M,t}$ is the 12-month change in the futures-based market reading and D_t is the FRB/US reading of the same asset changes less $I_{M,t}$. I call D_t the composition of the tightening: it is positive when the dollar accounts for more of a given tightening and stock prices for less. If the market reading is sufficient, $\gamma = 0$. If the FRB/US reading $I_{M,t} + D_t$ is sufficient, $\gamma = \beta$. The outcome closest to the spending calibration is real GDP growth over the next four quarters, since the FRB/US weights are effects on GDP at that horizon; I also report the change in the unemployment gap over the next 12 months.[3]

The test is demanding. The readings differ only when asset prices move in unusual proportions, so composition varies little once the market reading is held fixed: the part of composition unrelated to the market reading has about a fifth of the reading's own variation. Asset prices are also forward-looking. A fall in stock prices that reflects news about future earnings predicts weaker activity for reasons unrelated to the effect of wealth on spending, and a regression cannot tell the two apart. Indexes built to predict activity confirm the point from the other side: an index whose components are chosen to predict unemployment places sizable weight on equity prices and term spreads (Kiley, 2020), and conditions shift the lower tail of growth more than its center (Adrian, Boyarchenko and Giannone, 2019).

6.2 The Answer

For GDP growth, neither restriction is rejected in any sample (Table 5). The intervals are wide in economic terms: moving composition from its 5th to its 95th percentile at a given market reading changes predicted growth over the next year by 0.95 percentage point, with an interval from −0.85 to 2.76. Aggregate GDP is a blunt instrument for a question about channels. FRB/US gives the dollar a large weight because an appreciation reduces net exports, so the mix of asset movements should predict the mix of subsequent spending. Over 1976–2026 it does: at a given futures-based market reading, a higher FRB/US reading predicts a weaker contribution of net exports to growth over the next four quarters, with a coefficient of −0.74 (90 percent interval −1.06 to −0.41). The association runs mostly through equities, because rising stock prices strengthen domestic demand and imports; entered on its own, the dollar has the negative coefficient that the trade channel implies.

For unemployment, neither restriction is rejected over 1976–2026 or without 2008–15; the FRB/US restriction is rejected at the 10 percent level from 1996, and the sign of the composition

[3]The test compares two of the four calibrations. The other two are close to these two in the dimensions the test can detect.

Table 5: Does the Composition of a Tightening Help Predict Activity?

	n	Market reading, β	Market reading sufficient: $\gamma = 0$	FRB/US reading sufficient: $\gamma - \beta = 0$
Real GDP growth over the next four quarters				
1976–2026	186	−0.16 [−0.47, 0.16]	0.87 [−0.78, 2.51]	1.03 [−0.39, 2.44]
1996–2026	106	0.31 [−0.13, 0.75]	1.81 [−0.76, 4.38]	1.50 [−0.70, 3.71]
Excluding 2008–15	146	−0.33 [−0.61, −0.04]	0.26 [−1.21, 1.72]	0.58 [−0.69, 1.86]
Change in the unemployment gap over the next 12 months				
1976–2026	556	0.19 [0.05, 0.33]	−0.26 [−0.94, 0.43]	−0.45 [−1.01, 0.11]
1996–2026	318	0.01 [−0.21, 0.22]	−1.19 [−2.50, 0.13]	−1.19 [−2.31, −0.07]*
Excluding 2008–15	436	0.29 [0.15, 0.43]	0.32 [−0.13, 0.77]	0.03 [−0.30, 0.37]

Notes: Local projections of four-quarter-ahead real GDP growth (quarterly) and of the 12-month-ahead change in the unemployment gap (monthly) on own lags and $\beta M + \gamma D$, where M is the 12-month change of the futures-factor market reading and D the FRB/US reading of the same asset changes less M, both on the common scale. Brackets are 90 percent intervals (HAC, joint covariance of the coefficients); * marks a restriction rejected at the 10 percent level. If the market reading is a sufficient summary, $\gamma = 0$; if the FRB/US reading $M + D$ is, $\gamma = \beta$. The market weights are treated as known.

coefficient changes across the three samples.

Aggregate growth is too imprecise to rank the readings. The composition of a tightening does predict the composition of subsequent spending, through net exports, and in this sample mostly through equities and imports.

7 Scope and Conclusion

7.1 Scope

The four calibrations are a defined set of U.S. calibrations on three asset prices. Credit spreads matter for monetary transmission: Gertler and Karadi (2015) find that policy surprises move credit costs, including the excess bond premium of Gilchrist and Zakrajšek (2012), through term and credit premiums, and Caldara and Herbst (2019) find that policy responds to credit spreads and that leaving spreads out of a VAR distorts the estimated effects of policy. The 10-year yield and equity valuation carry a sizable part of that information. Over 1996–2026 the three asset prices of the paper account for 39 percent of the variance of 12-month changes in the excess bond premium, mostly through equity valuation. Spreads, house prices and the other variables that some indexes include are outside the comparison of weights; adding spreads would raise every reading in 2008. Only the market weights are estimated here. The intervals come from resampling the 253 announcements and treat the FRB/US and GS weights, the observed history and r^* as given. The market weights are estimated on announcements since 1996 and held fixed. Re-estimated on an expanding window, the weights on the 10-year yield and the dollar stay within the sampling error of the full-sample estimates, and the weight on equities lies just outside it in estimates between 2011 and early 2015 (Appendix C.3). If policy came to target financial conditions, the responses on which the market weights rest could change, as Hatzius et al. (2010) note for all such indexes.

7.2 Conclusion

What can a policymaker infer from a report that financial conditions have tightened by the equivalent of a quarter point? The four calibrations compared here usually agree on the direction of a change and often disagree about its size. The size depends on two choices, the weights that turn movements in asset prices into points of policy and the neutral level against which the asset prices are measured. In 2008–09 each choice moved the reading by more than half a point.

In this sample, knowing how an index was built says little about which other index it will resemble: the futures-based market calibration and the published GS weights give the closest readings, although one is built from policy news and the other from spending effects. In a market calibration, a large equity weight means that stock prices help identify policy news. In a spending calibration, it means that stock prices have a large estimated effect on demand. The futures-based and GS calibrations arrive at similar equity and dollar weights by answering those different questions. On the neutral level, measuring relative to r^* tracks the gap to neutral exactly, up to a constant, when the weighted neutral premiums are constant.

A reported tightening should therefore state three things: the asset prices that moved, the neutral benchmark against which they were measured, and the economic meaning assigned to the weights. That is also the question an analysis of targeting financial conditions, such as Caballero and Simsek (2026), has to answer before a point of the target can be given a meaning.

References

Acosta, M., A. Ajello, M. Bauer, F. Loria, and S. Miranda-Agrippino. 2025. "Financial Market Effects of FOMC Communication: Evidence from a New Event-Study Database." *Federal Reserve Bank of San Francisco Working Paper 2025-30.*

Adrian, T., N. Boyarchenko, and D. Giannone. 2019. "Vulnerable Growth." *American Economic Review*, 109(4): 1263–1289.

Ajello, A., M. Cavallo, G. Favara, W. Peterman, J. Schindler, and N. Sinha. 2023. "A New Index to Measure U.S. Financial Conditions." *FEDS Notes, Board of Governors of the Federal Reserve System, June 30.*

Aramonte, S., S. Rosen, and J. W. Schindler. 2013. "Assessing and Combining Financial Conditions Indexes." *Finance and Economics Discussion Series 2013-39, Board of Governors of the Federal Reserve System.*

Arnaut, Z., and M. D. Bauer. 2024. "Monetary Policy and Financial Conditions." *Federal Reserve Bank of San Francisco Economic Letter 2024-07, March 4.*

Bauer, M. D., and E. T. Swanson. 2023*a*. "An Alternative Explanation for the 'Fed Information Effect'." *American Economic Review*, 113(3): 664–700.

Bauer, M. D., and E. T. Swanson. 2023*b*. "A Reassessment of Monetary Policy Surprises and High-Frequency Identification." *NBER Macroeconomics Annual 2022*, 37: 87–155.

Bernanke, B. S., and K. N. Kuttner. 2005. "What Explains the Stock Market's Reaction to Federal Reserve Policy?" *Journal of Finance*, 60(3): 1221–1257.

Bletzinger, T., G. Martorana, and J. Mistak. 2026. "The Evolution of Financial Conditions in the Euro Area and the United States: A Macro-Finance FCI Perspective." *ECB Economic Bulletin, Issue 6/2026, Box.*

Boehm, C. E., and T. N. Kroner. 2025. "Monetary Policy without Moving Interest Rates: The Fed Non-Yield Shock." *NBER Working Paper 32636 (June 2024; revised September 2025).*

Brave, S., and R. A. Butters. 2011. "Monitoring Financial Stability: A Financial Conditions Index Approach." *Federal Reserve Bank of Chicago Economic Perspectives*, 35(1): 22–43.

Brayton, F., T. Laubach, and D. Reifschneider. 2014. "The FRB/US Model: A Tool for Macroeconomic Policy Analysis." *FEDS Notes, Board of Governors of the Federal Reserve System, April 3.*

Caballero, R. J., and A. Simsek. 2026. "Financial Conditions Targeting in a Multi-Asset Open Economy." *NBER Working Paper 34974.*

Caballero, R. J., T. E. Caravello, and A. Simsek. 2024. "Financial Conditions Targeting." *NBER Working Paper 33206.*

Caballero, R. J., T. E. Caravello, and A. Simsek. 2026. "FCI-star." *NBER Working Paper 33952 (version of January 12, 2026).*

Caldara, D., and E. Herbst. 2019. "Monetary Policy, Real Activity, and Credit Spreads: Evidence from Bayesian Proxy SVARs." *American Economic Journal: Macroeconomics*, 11(1): 157–192.

Dudley, W., and J. Hatzius. 2000. "The Goldman Sachs Financial Conditions Index: The Right Tool for a New Monetary Policy Regime." *Goldman Sachs Global Economics Paper No. 44, June 8.*

Eika, K. H., N. R. Ericsson, and R. Nymoen. 1996. "Hazards in Implementing a Monetary Conditions Index." Board of Governors of the Federal Reserve System International Finance Discussion Paper 568. doi:10.17016/ifdp.1996.568. Published in Oxford Bulletin of Economics and Statistics 58(4), 765–790, doi:10.1111/j.1468-0084.1996.mp58004009.x.

Gertler, M., and P. Karadi. 2015. "Monetary Policy Surprises, Credit Costs, and Economic Activity." *American Economic Journal: Macroeconomics*, 7(1): 44–76.

Gilchrist, S., and E. Zakrajšek. 2012. "Credit Spreads and Business Cycle Fluctuations." *American Economic Review*, 102(4): 1692–1720.

Gürkaynak, R. S., B. Sack, and E. T. Swanson. 2005. "Do Actions Speak Louder Than Words? The Response of Asset Prices to Monetary Policy Actions and Statements." *International Journal of Central Banking*, 1(1): 55–93.

Hatzius, J., and S. J. Stehn. 2018. "The Case for a Financial Conditions Index." *Goldman Sachs Global Economics Paper, July 16.*

Hatzius, J., P. Hooper, F. Mishkin, K. Schoenholtz, and M. Watson. 2010. "Financial Conditions Indexes: A Fresh Look after the Financial Crisis." *NBER Working Paper 16150; also in U.S. Monetary Policy Forum Report*, 2010: 3–59.

Holston, K., T. Laubach, and J. C. Williams. 2017. "Measuring the Natural Rate of Interest: International Trends and Determinants." *Journal of International Economics 108, S59–S75.*

Holston, K., T. Laubach, and J. C. Williams. 2023. "Measuring the Natural Rate of Interest after COVID-19." *Federal Reserve Bank of New York Staff Report 1063.*

Kiley, M. T. 2020. "Financial Conditions and Economic Activity: Insights from Machine Learning." *Finance and Economics Discussion Series 2020-095, Board of Governors of the Federal Reserve System.*

Kuttner, K. N. 2001. "Monetary Policy Surprises and Interest Rates: Evidence from the Fed Funds Futures Market." *Journal of Monetary Economics*, 47(3): 523–544.

Laubach, T., and J. C. Williams. 2003. "Measuring the Natural Rate of Interest." *Review of Economics and Statistics*, 85(4): 1063–1070.

Nakamura, E., and J. Steinsson. 2018. "High-Frequency Identification of Monetary Non-Neutrality: The Information Effect." *Quarterly Journal of Economics*, 133(3): 1283–1330.

Powell, J. H. 2023. "Transcript of Chair Powell's Press Conference." *Board of Governors of the Federal Reserve System, November 1.*

Powell, J. H. 2024. "Transcript of Chair Powell's Press Conference." *Board of Governors of the Federal Reserve System, March 20.*

Sack, B., and E. Swanson. 2023. "The Impact of Federal Reserve Policy on the Fed's Financial Conditions Index." *Brookings Commentary, October 18.*

Appendix A The Weights: a Derivation

Let the movements in the three asset prices be $x = \psi s + u$, where s is policy news, ψ the responses and u everything else, with covariance Σ. Let g_h be the effects of a persistent unit shift in each asset price on spending at horizon h. The FRB/US weights are proportional to g_h. The market weights solve the problem of recovering s from x: the generalized least squares estimator is $\hat{s} = w_M' x$ with $w_M = \Sigma^{-1}\psi/(\psi'\Sigma^{-1}\psi)$, which satisfies $w_M'\psi = 1$ and, by the Gauss–Markov theorem, minimizes the variance of the estimate among linear unbiased estimators of s when u has mean zero and covariance Σ and is uncorrelated with s. For any x the difference between the normalized readings is $D_h(x) = (w_{B,h} - w_M)'x = (w_{B,h} - w_M)'u$, because the policy part cancels. The readings coincide for every movement if and only if $w_{B,h} = w_M$, that is, if $g_h \propto \Sigma^{-1}\psi$: the criterion that recovers policy news precisely and the criterion that measures spending effects select the same weights only in that case. The variance of the difference is $(w_{B,h} - w_M)'\Sigma(w_{B,h} - w_M)$.

Appendix B Construction of the Series

The three asset prices enter in three forms, each answering a problem of the one before. The headline results use the third.

Nominal. Rates as published. A nominal index treats a 10-year yield of 8 percent alike in 1981 and in 2000. That is reasonable only if inflation expectations and the neutral real rate are stable, and neither was. Expected inflation over the next year averaged 8.43 percent in 1981 and is 2.66 percent today, and the Laubach–Williams estimate of r^* averaged 2.76 percent over 1990–2007 and 0.86 percent over 2009–2016 (Figure B.1).

Real. Each rate less expected inflation at a matched horizon: ten-year expected inflation for the 10-year yield. The dollar is the real broad index.

Relative to r^*. The real 10-year yield less r^*. Equity valuation enters through the earnings yield $E/P = 100/\text{CAPE}$, which is also an implicit real rate, as the gap $E/P - r^*$, converted to CAPE log points at the sample-average earnings yield so that its weight on CAPE is unchanged. The baseline uses the one-sided, full-sample Laubach–Williams estimate. Each variable is measured relative to its 1973–2026 average, so r^* matters through its movements over time.

Data. All series are monthly averages from January 1973 to September 2026, from the public sources in Table B.1. The BBB spread is Moody's Baa yield less the 10-year yield. Months in which the CPI and unemployment releases are missing (October 2025) are interpolated. The unemployment gap is the unemployment rate less the Congressional Budget Office's natural rate, placed in the middle month of each quarter and interpolated. Expected inflation is the Cleveland Fed's estimate at one and ten years from 1982 and the Livingston Survey's 12-month CPI expectation before, shifted to the Cleveland level by the mean difference over 1982–84 and blended over 1982. CAPE is Shiller's series spliced over 1990–92 in logs to the monthly average of a daily series. The real broad dollar is spliced in 2006 in logs, as are the nominal dollar and the S&P 500.

Market index. Policy news is the first principal component of MP1, MP2, ED2, ED3 and ED4

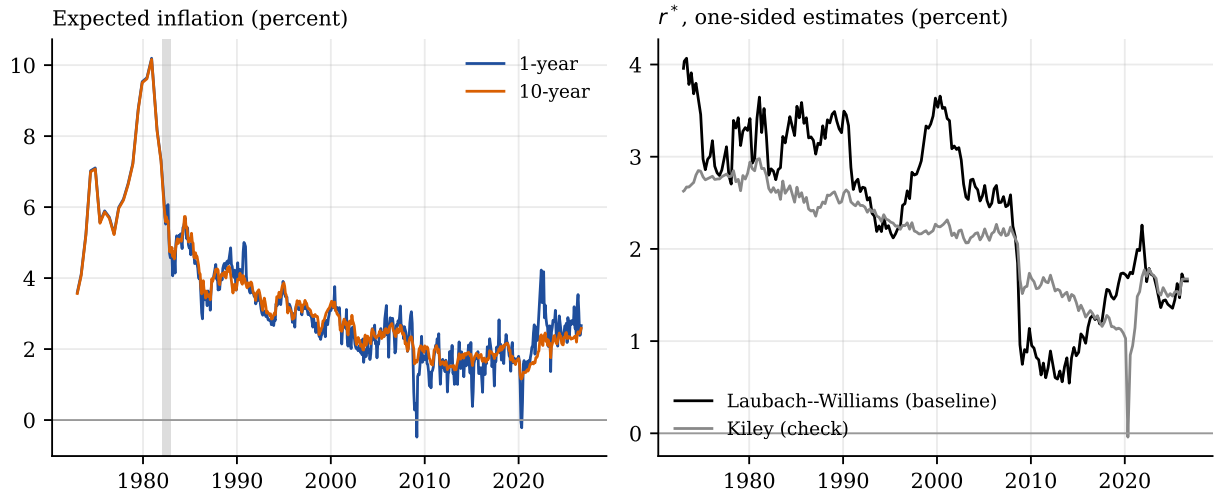


Figure B.1: Expected Inflation and r^*

Notes: Left: expected inflation, Livingston Survey before 1982 and Cleveland Fed model estimates after, shifted to the Cleveland level over 1982–84 and blended over 1982; shaded: the blending year. Right: one-sided, full-sample estimates of r^* from the Laubach–Williams model (baseline) and Kiley's model (check). Percent.

in the monetary-event windows, scaled by its regression on the two-year yield change so that one unit moves the two-year yield by one percentage point. Unscheduled events with no change in MP1 and March–April 2020 are dropped. Each of the 10-year yield change, the S&P 500 return and the dollar (DXY) return in the window is regressed on the factor. The residual covariance Σ of the three regressions gives $w = \Sigma^{-1}\psi/(\psi'\Sigma^{-1}\psi)$. The S&P 500 and DXY weights apply to 100 times the log of CAPE and of the real broad dollar, as percent changes in the window and in the index levels agree to first order. The weights bootstrap resamples announcements with replacement and recomputes the policy-news factor and the weights each time; draws use the seed in `lib/config.py`.

FRB/US index. The weights are the cumulative GDP responses at four quarters to a persistent unit shift in each variable, simulated in FRB/US with VAR-based expectations, divided by the response to a persistent one percentage point rise in the funds rate. The weight on the policy rate is its direct effect with expectations of future rates held fixed. House prices enter as 100 times the log of the price-to-rent ratio relative to its trailing 15-year average. The expenditure decomposition of Figure 3 uses the model's own identities for the components of GDP, weighted by their shares; it is run with the Board's FRB/US package and the data vintage of October 2026, downloaded by step 00 of the replication package.

Policy-news definitions. The two-year definition replaces the futures factor by the change in the two-year Treasury yield in the same window, with the same regressions and the same bootstrap of announcements. The Bauer–Swanson definition uses their surprise orthogonalized to public news (Bauer and Swanson, 2023*b*), matched to this paper's announcements through 2019 and scaled so that one unit raises the two-year yield by one percentage point. The announcement bootstrap uses

Table B.1: Data Sources

Series	Frequency	Source
Federal funds rate; 10-year yield; Baa yield; 30-year mortgage rate	Monthly	FRED FEDFUNDS, GS10, BAA, MORTGAGE30US
Real broad dollar; nominal broad dollar	Monthly	FRED RTWEXBGS, TWEXBPA; DTWEXBGS, TWEXBMTH (spliced in 2006)
CPI; CPI rent; Case–Shiller home prices; S&P 500	Monthly	FRED CPIAUCSL, CUUR0000SEHA, CSUSHPISA, SP500; Shiller's data before the FRED start
Unemployment rate; natural rate of unemployment; real GDP	Monthly; quarterly	FRED UNRATE; CBO, FRED NROU; GDPC1
Expected inflation, 1 and 10 years	Monthly	Cleveland Fed, FRED EXPINF1YR, EXPINF10YR; before 1982, Livingston Survey (Philadelphia Fed)
r^*, Laubach–Williams	Quarterly	New York Fed, one-sided estimates
r^*, Kiley	Quarterly	One-sided estimates of the author
CAPE	Monthly	Shiller, spliced to the monthly average of a daily CAPE constructed by the repository's daily workflow
FRB/US simulations	Quarterly	Run for this paper's companion work with the public FRB/US package
Futures, yields, S&P 500 and DXY in the window	Event	U.S. Monetary Policy Event-Study Database, "monetary events" (Acosta et al., 2025)

5,000 draws for every weights-based interval; the block bootstraps of the outcome regressions use 500.

Forecasting data. The regressions use local projections of the 12-month-ahead change in the unemployment gap (four-quarter-ahead GDP growth) with HAC standard errors of 18 lags.

Appendix C The Market Weights

C.1 Estimates and Precision

Table C.1 gives the market weights under the baseline. A policy move that raises the two-year yield by one point raises the 10-year yield by 0.53 points, lowers equity valuation by 7.78 percent and raises the real dollar by 4.22 percent. Each asset carries a substantial part of the policy move, with shares of 0.33 for the 10-year yield, 0.40 for equity valuation and 0.28 for the dollar. The 90 percent intervals of the weights, from 5,000 bootstrap draws of the 253 announcements, run from 0.15 to 1.11 for the 10-year yield, -0.07 to -0.03 for equity valuation and 0.02 to 0.11 for the dollar.

C.2 Other Measures of Policy News and Event Samples

Table C.2 re-estimates the weights with the two-year yield and the Bauer–Swanson orthogonalized surprise as the measure of policy news, with announcements through 2019, and without the lower-

Table C.1: Market Index: Responses to Policy News and Weights

	Response	Weight	90% interval	Share of policy move
10-year Treasury yield (pp)	0.53 (0.06)	0.609	[0.147, 1.114]	0.33
Equity valuation, 100 ln CAPE (percent)	−7.78 (0.93)	−0.051	[−0.068, −0.030]	0.40
Real broad dollar, 100 ln (percent)	4.22 (0.53)	0.065	[0.021, 0.112]	0.28

Notes: Announcement windows, 253 announcements, January 1996 to September 2026. Response: change in the window per unit of policy news, which moves the two-year yield by 1 percentage point (heteroskedasticity-robust standard errors in parentheses). Weight: one-factor generalized least squares, normalized so that the weights applied to the responses sum to 1. Interval: 5th and 95th percentiles of weights re-estimated on 5000 bootstrap samples of the announcements. Share: weight times response.

Table C.2: Weights Under Other Policy-News Series and Event Samples

	Events	w 10-year	w equities	w dollar	ψ 10-year	ψ equities	ψ dollar
Baseline: Nakamura–Steinsson factor	253	0.609	−0.051	0.065	0.53	−7.78	4.22
Policy news is the 2-year yield change	253	0.925	−0.027	0.070	0.58	−6.09	4.22
Announcements through 2019	201	0.613	−0.061	0.053	0.56	−7.50	3.83
No announcements December 2008–November 2015	197	0.986	−0.039	0.058	0.49	−7.82	3.69
Policy news is the Bauer–Swanson orthogonalized surprise (to 2019)	200	0.542	−0.070	0.040	0.58	−8.93	3.98

Notes: Weights w and responses ψ as in Table C.1, re-estimated on each sample or with another policy-news measure: the window change of the two-year yield, or the surprise of Bauer and Swanson (2023*b*) orthogonalized to public news before the announcement, matched to this paper's announcements through 2019 and scaled so that one unit raises the two-year yield by one percentage point. In each case the weights applied to the responses sum to 1.

bound years. The equity weight is −0.027, −0.061 and −0.039 under the two-year yield and the two sample checks, and −0.070 under the Bauer–Swanson surprise, against −0.051 in the baseline. FRB/US gives the dollar more weight relative to equities than every one of these market calibrations.

C.3 Weights That Move

The weights define the unit, so I hold them fixed and show how they would drift. Re-estimating on an expanding window of announcements from 2006 keeps the 10-year and dollar weights inside the 90 percent band of the full-sample estimate at every date after 2010; the equity weight is outside the full-sample 90 percent bootstrap interval in 32 monthly estimates between January 2011 and January 2015, by at most −0.073 against a band edge of −0.068 (Figure C.1). The overlapping monthly estimates are not 32 independent tests of instability. A 20-year rolling window fails after 2021, because it drops the announcements of 1996–2005. The 10-year weight is imprecise throughout. Apart from the equity weight in these years, the weights stay within their sampling error.

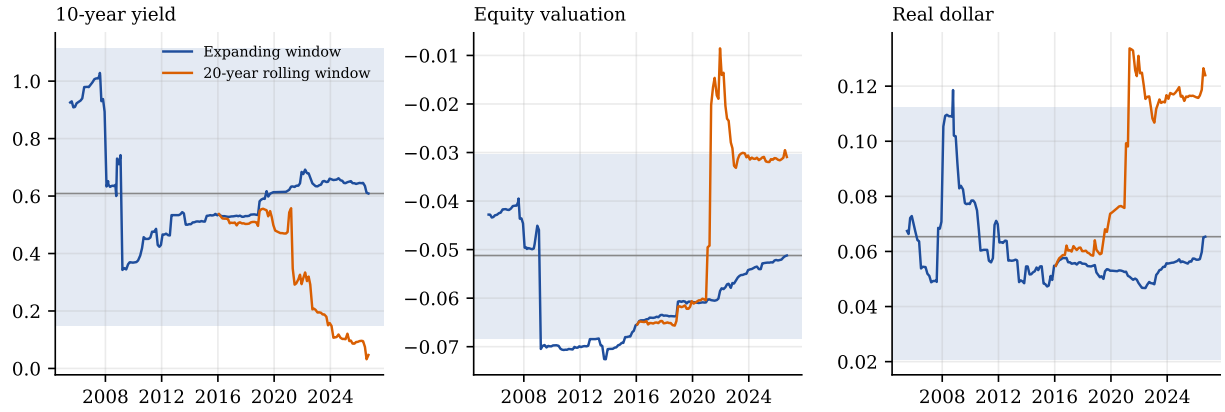


Figure C.1: The Weights on an Expanding and a Rolling Window

Notes: Weights of the market index re-estimated on announcements to each date (expanding window, from the 80th announcement) and on the latest 20 years (rolling). Shaded: 90 percent bootstrap interval of the full-sample weights; the horizontal line is the full-sample weight.